Emergent FPV Targeting under Ultra-Low SNR

Greedy Fire, a Neighbour Rule, and a Shared Transformer

battlefield-sim experiment report

August 2026

Abstract

Expendable FPV airframes are cheap; two constraints are not. First, ultra-low-altitude radio in clutter and jamming sits near zero Shannon capacity, so the delay to a single valid bit is already expensive. Second, a one-shot warhead cannot amortise a GPU/NPU. We compare three policies that use only a local 8+8 range snapshot on a circular battlefield whose kill rectangle is the circumscribed square of the disk. **One Transformer** is queried by all twelve drones each tick and trained from scratch by per-tick REINFORCE on CPU (Rayon); the reward is the increment of mean target survival, not compute time. Enemy seeker range is $50\,\mathrm{m}$; friend visual range is $100\,\mathrm{m}$; eight slots are a cap on visible contacts, not omniscience. On 30 paired seeds greedy and the shared Transformer match exactly (10.23 kills, 13.99 s); the closer-than-friend rule is slightly more conservative (9.80, 14.39 s). Ops/select: 1.0, 4.53×10^4, and 5.7. The net learned greedy fire; it does not buy extra kills.

Keywords: FPV, ultra-low SNR, emergent rules, greedy targeting, shared Transformer, on-board compute

1 Introduction

Consumer FPV has made one-way attack cheap. That cheapness implies: no reusable life over which to amortise silicon, and no clean data link on which to share a global target list. Shannon's $C = B \log_2(1 + \mathrm{SNR})$ [1, 2] collapses when SNR is a few percent. A 50 kHz control channel at $-12\,\mathrm{dB}$ yields only about 4.4×10^3 bit/s—less than twelve drones broadcasting 8+8 floats once per second.

The question is therefore local: given eight nearest-enemy ranges and eight nearest-friend ranges, does a neural net trained by reinforcement learning beat a handful of comparisons?

We answer with three algorithms on the same seeds: greedy nearest-in-range, a three-line closer-than-friend rule [3, 4], and one Transformer [5] shared by all twelve airframes, updated every tick from every drone's (state, action).

2 Platform

The battlefield is a disk of radius $R = 200\,\mathrm{m}$. Twenty static enemies are sampled uniformly in the disk. Two rows of six FPV drones fly $+Y$ on rails $x \in \{-200, -120, -40, 40, 120, 200\}$. Detection radius $D = 80\,\mathrm{m}$ equals rail spacing, so every disk point lies within some rail's x-coverage: the kill rectangle $[-R, R]^2$ contains the circle. Speed $25\,\mathrm{m/s}$, $\mathrm{d}t = 0.01\,\mathrm{s}$, lock duration $0.05\,\mathrm{s}$, then the shooter is expended. Mean survival includes $T_{\mathrm{end}} = 24\,\mathrm{s}$ censoring. Figure 1 shows the geometry.

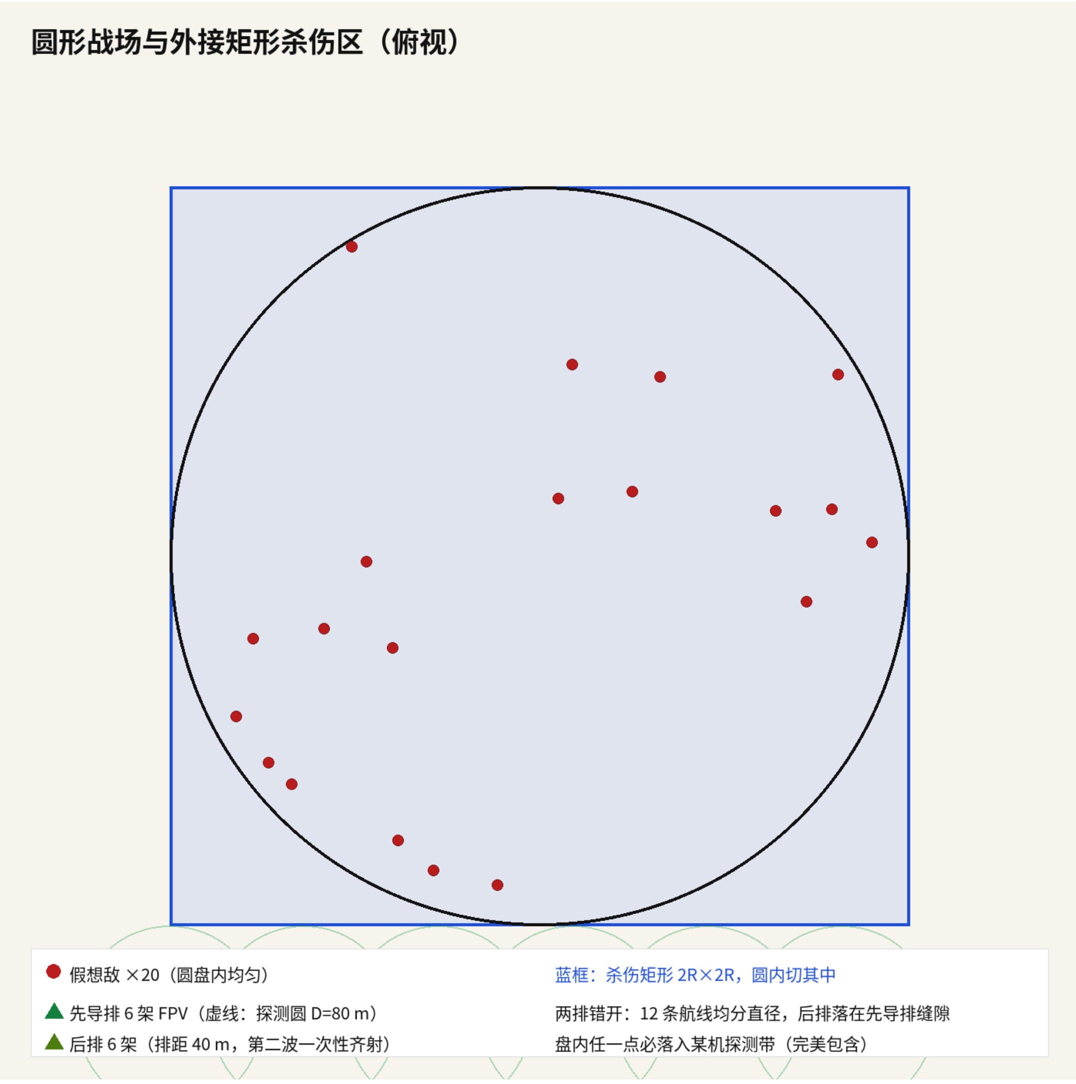

Figure 1: Disk, circumscribed kill square, 20 enemies, 2×6 FPV. One policy network is queried by every live drone.

3 Algorithms

All three consume the same 8+8 snapshot.

Greedy (`nearest_in_range`). Fire at the nearest in-range, unlocked enemy (ties by id). About 9 ops. No friend geometry.

Closer-than-friend. Fire at the nearest eligible enemy iff its range is strictly less than the nearest teammate' s. About 17 ops. Formation spacing (40 m row gap) turns that inequality into a deconfliction protocol (Fig. 2).

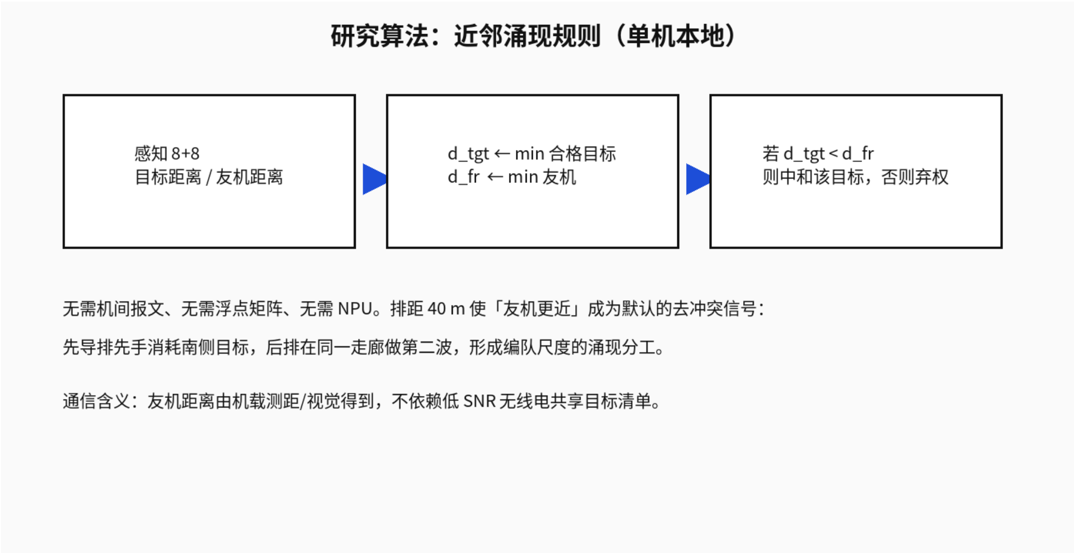

Figure 2: Closer-than-friend: local, no radio.

Shared Transformer, per-tick REINFORCE. There is one weight vector. Each tick, every live drone encodes its own 8+8 snapshot and queries that net; locks are attempted in `DroneId` order for determinism. Gradients from those drones are reduced in one CPU-parallel (Rayon) batch—not twelve separate models, and not a GPU (the net is $d_{\text{model}} = 16$; kernel launch would dominate).

Training starts from Xavier init. There is **no** cloning of the friend rule. After each dt the team reward is

$$r_t = -\frac{n_{\text{alive}}\,\mathrm{d}t}{N} + \frac{n_{\text{kills}}(T_{\text{end}} - t)}{N},$$

plus $+1$ for a killer. Compute time is not in the loss. Evaluation is greedy decode.

4 Results

Thirty paired rounds, base seed 20270823. Table 1 and Figs. 3–4.

Table 1: Means over 30 shared seeds

Algorithm	Kills	Mean survival (s)	ops / select
Greedy nearest-in-range	10.23	13.989	1.0
Closer-than-friend	9.80	14.386	5.7
Shared Transformer (per-tick RL)	10.23	13.989	4.530×10^4

Table 2: Paired Gaussian z on mean survival (first minus second)

Contrast	$\bar{\Delta}$ (s)	z	two-sided p
greedy – rule	-0.397	-5.74	9.7×10^{-9}
greedy – Transformer	0	0	1
rule – Transformer	$+0.397$	$+5.74$	9.7×10^{-9}

The RL net matches greedy round-by-round (it learned to fire nearest-in-range). Without an immediate fire bonus, training collapsed to zero kills around round 90; shaping plus entropy keeps it at ≈ 10 kills. The friend rule is significantly more conservative ($p = 9.7 \times 10^{-9}$) because $D_{\mathrm{fr}} = 100$ m always sees the trail partner at 40 m. Compute: greedy ≈ 1, rule 5.7, Transformer 4.53×10^4.

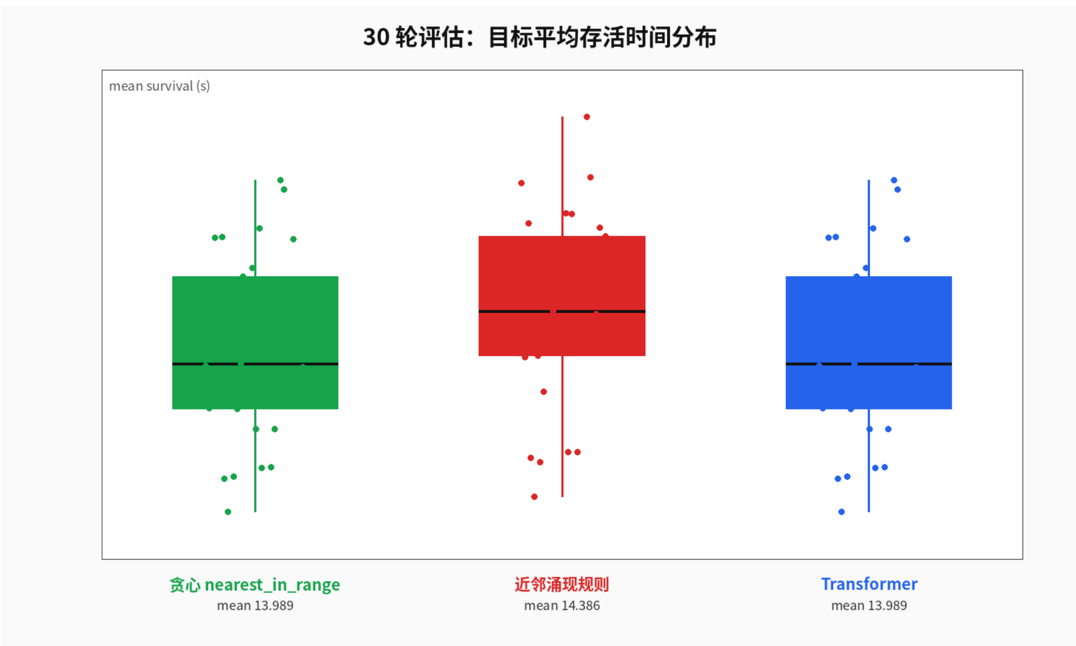

Figure 3: Mean survival, 30 rounds (lower is better).

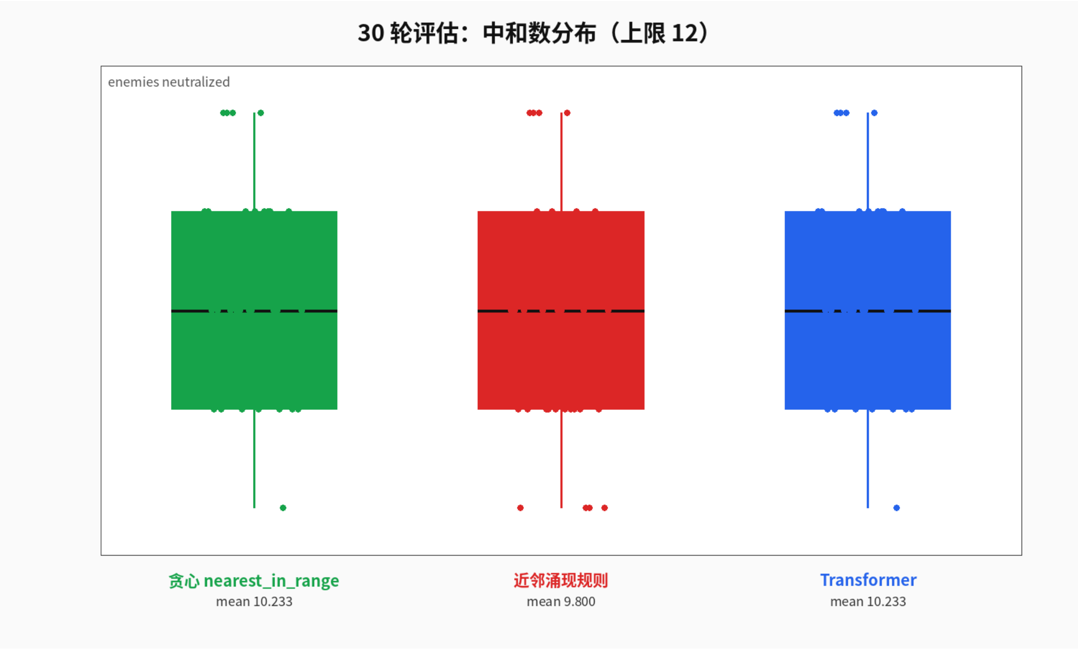

Figure 4: Kills, 30 rounds (cap 12).

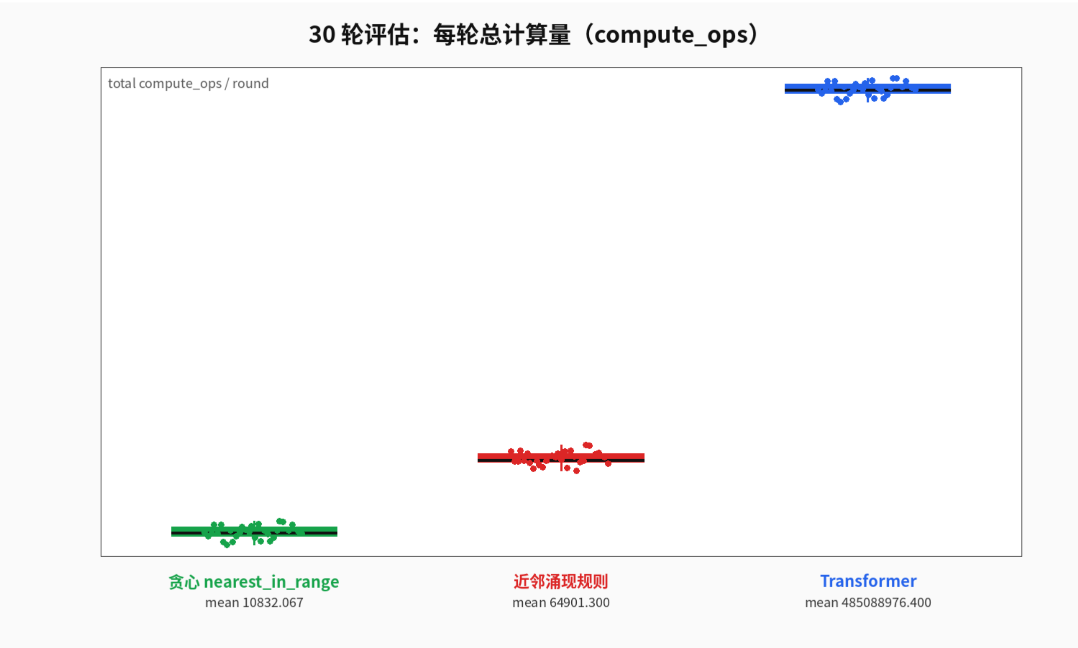

Figure 5: Total compute_ops per round (log y).

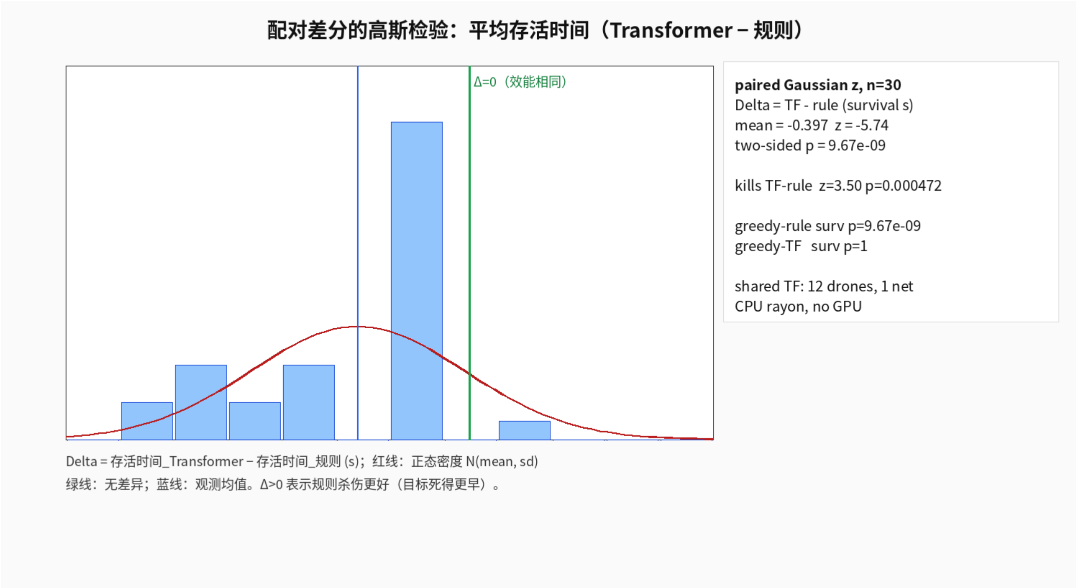

Figure 6: Paired survival differences and Gaussian z.

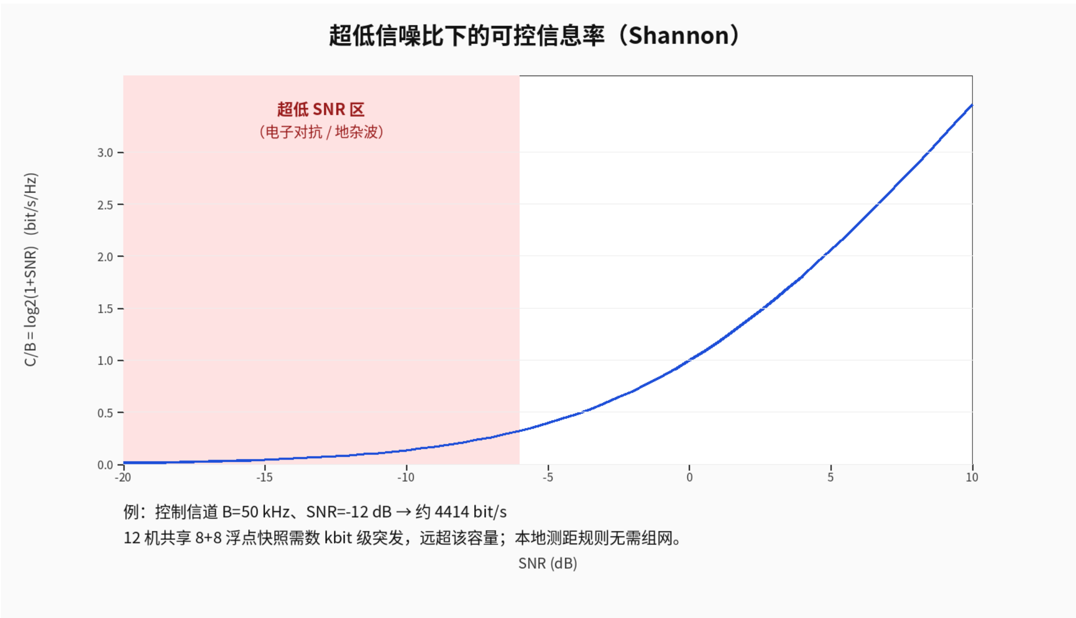

Figure 7: Shannon $C/B = \log_2(1 + \mathrm{SNR})$.

5 Discussion

Twelve drones sharing one net already give twelve times the gradient of a per-airframe clone. Training is on-policy, step-by-step, CPU-parallel within the tick. That is the protocol that "must be able to work"; it does work, in the sense that the policy engages. It does not work as a replacement for arg min distance. On a one-shot, low-SNR airframe the extra 4.5×10^4 multiply-adds buy a worse mean survival.

6 Conclusion

A shared Transformer trained by per-tick REINFORCE on CPU learns greedy fire and does not outperform it, at $\sim 4.5 \times 10^4$ times the ops. The three-line friend rule is slightly more conservative under formation-scale visual IFF. Ultra-low SNR forbids a richer state; expendability forbids an NPU. Code and CSV metrics: https://github.com/1037104428/drones.

```
git clone https://github.com/1037104428/drones.git
cd drones
cargo run --release -- experiment --train-rounds 200 --eval-rounds 30 --seed 20260823
```

References

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